



Redmi S2 Level 3 Maintenance Guide V01

Technical support internal document control: TSIMHNE6 Redmi S2 Level 3 Maintenance Guidance V01

Scope of application:

Analysis center, mainboard and whole machine maintenance factory

Change history:

First edition, August 20, 2018

- Redmi S2 Level 3 Maintenance Guidance
 - V01 1. Introduction of Basic Information
 - [1.1](#) Product Overview
 - [1.2](#) Special welding fixture for red rice S2
 - [1.3](#) Redmi S2 power supply adapter
 - [1.4](#) Pasting position and specification of maintenance label
 - [1.5](#) Matters needing attention in motherboard maintenance
 - [1.6](#) Brush mode
 - [1.7](#) RF calibration test correlation
 - [2](#). Brief introduction of motherboard module
 - [2.1](#) Redmi S2 motherboard component distribution diagram
 - [2.2](#) Brief introduction of red rice S2
 - startup timing and key signal measurement table 3. Troubleshooting
 - [3.1](#) Switching failure
 - [3.1.1](#) Constant current without startup
 - [3.1.2](#) No startup current is not maintained
 - [3.1.3](#) No startup, no current
 - [3.1.4](#) Leakage without starting up
 - [3.2](#) Restart failure
 - [3.3](#) Crash failure
 - [3.4](#) Signal fault
 - [3.5](#) SIM card failure
 - [3.6](#) Charging failure
 - [3.7](#) Display fault
 - [3.8](#) Audio fault
 - [3.8.1](#) Loudspeaker failure

- [3.8.2](#) MIC fault
- [3.8.3](#) Receiver failure
- [3.8.4](#) Headphone failure
- [3.9](#) WIFI/BT/FM/GPS Fault
- [3.10](#) Camera failure
- [3.11](#) Inductor failure
- [3.12](#) Touch screen failure
- [3.13](#) Fingerprint identification failure

1. Introduction of basic information

1.1 Product Overview

Product Overview:



红米自拍手机 S2

前置1600万智能美拍
2 μ m大像素 (4合1)

后置1200万+500万
AI 双摄，拍照更出色

AI 人脸解锁
看一眼就解锁

5.99"全面屏
18:9 电影般的视觉体验

骁龙625 八核处理器
14nm工艺 高性能低功耗

2+1 卡槽
最高可扩展256GB



玫瑰金



香槟金



铂银灰

网络与制式

2 + 1 卡槽

双Nano-Sim卡+micro-SD拓展

网络频段

详细网络频段

GSM (频段 B2/3/5/8)

CDMA (频段 BC0)

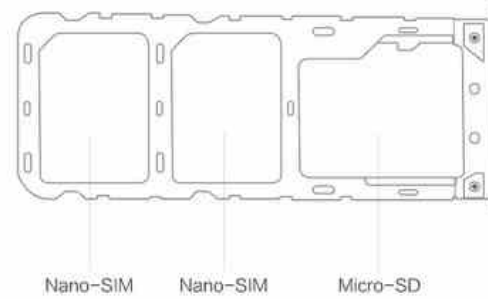
CDMA EVDO (频段 BC0)

WCDMA (频段B1/2/5/8)

TD-SCDMA (频段 B34/39)

TDD-LTE (频段 B34/B38/39/40/41 (100MHz))

FDD-LTE (频段 B1/3/5/7/8)



双卡使用说明

支持移动、联通、电信 4G/3G/2G，支持移动、联通带内载波聚合4G+网络

当主卡为移动、联通4G时，副卡支持联通3G网络语音电话

注：两张电信卡不能同时使用

无线网络

支持802.11 b/g/n WIFI协议

蓝牙4.2 无线技术

WiFi Display, WiFi Direct

1.2 Special welding fixture for red rice S2

Material code: SCNC030039100



The fake wire of Redmi S2 is the same as that of Xiaomi MI5X. Material code: SCNC030039100



1.4 Pasting position and specification of maintenance label

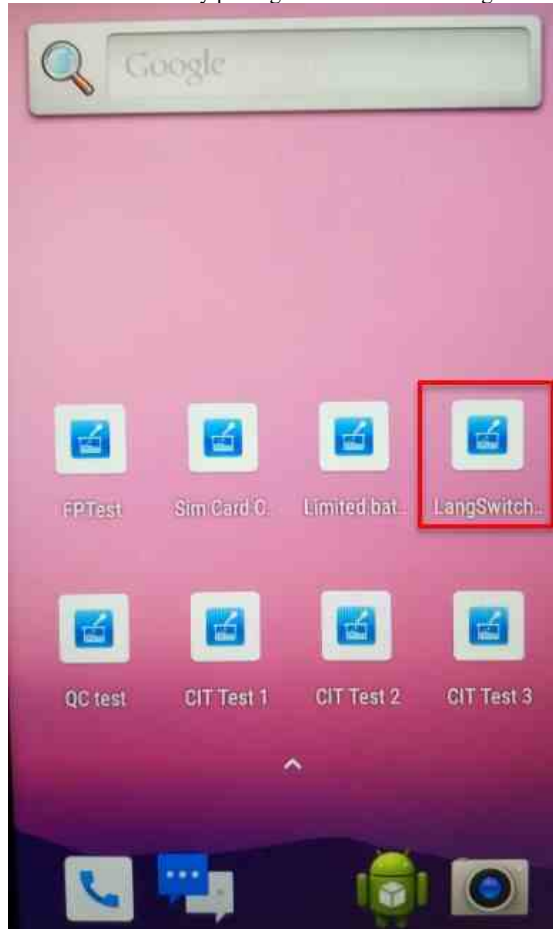
Redmi S2
Maintenance Label:
The pasting position
is as shown in the
figure:



1.5 Matters needing attention in motherboard maintenance

Note:

1. After replacing EMMC, it can be turned on normally in factory mode. If TEEKEY is not uploaded, it will be turned on and restarted in user mode.
2. Brush in the factory package and boot it to the English interface by default. You can click "LangSwitch" on the desktop to switch between Chinese and English.



is near the key interface, display interface and earphone interface. These plastic components are easy to weld.

1.6 Brush mode

Brush platform: MiFlash

Special factory package for changing number (keep NV package):
E6_FACTORY_LLDLMC9_T519_0424_KeepNV
special factory package for
maintenance (clear NV package): E6_FACTORY_LLDLMC9_T519_0424_eraseNV
brush mode: deep brush

1. When the mobile phone is turned off, short-circuit TP2703 and TP2704 to make the mobile phone enter deep brush mode (identify port 9008).



2. Select the corresponding software package, check all delete options at the bottom of MiFlash, click Load Device, and click Brush to log in to Xiaomi account to start brushing.

选择

D:\E6\工厂包\E6_FACTORY_LLDMC9_T519_0424_keepNV

加载设备

刷机

编号

4

刷机工具

English

账号登录成功

当前电脑已绑定，可以开始刷机

退出登录



213005688

解绑电脑

☒ 全部删除 ☐ 保留用户数据 ☐ 全部删除并lock



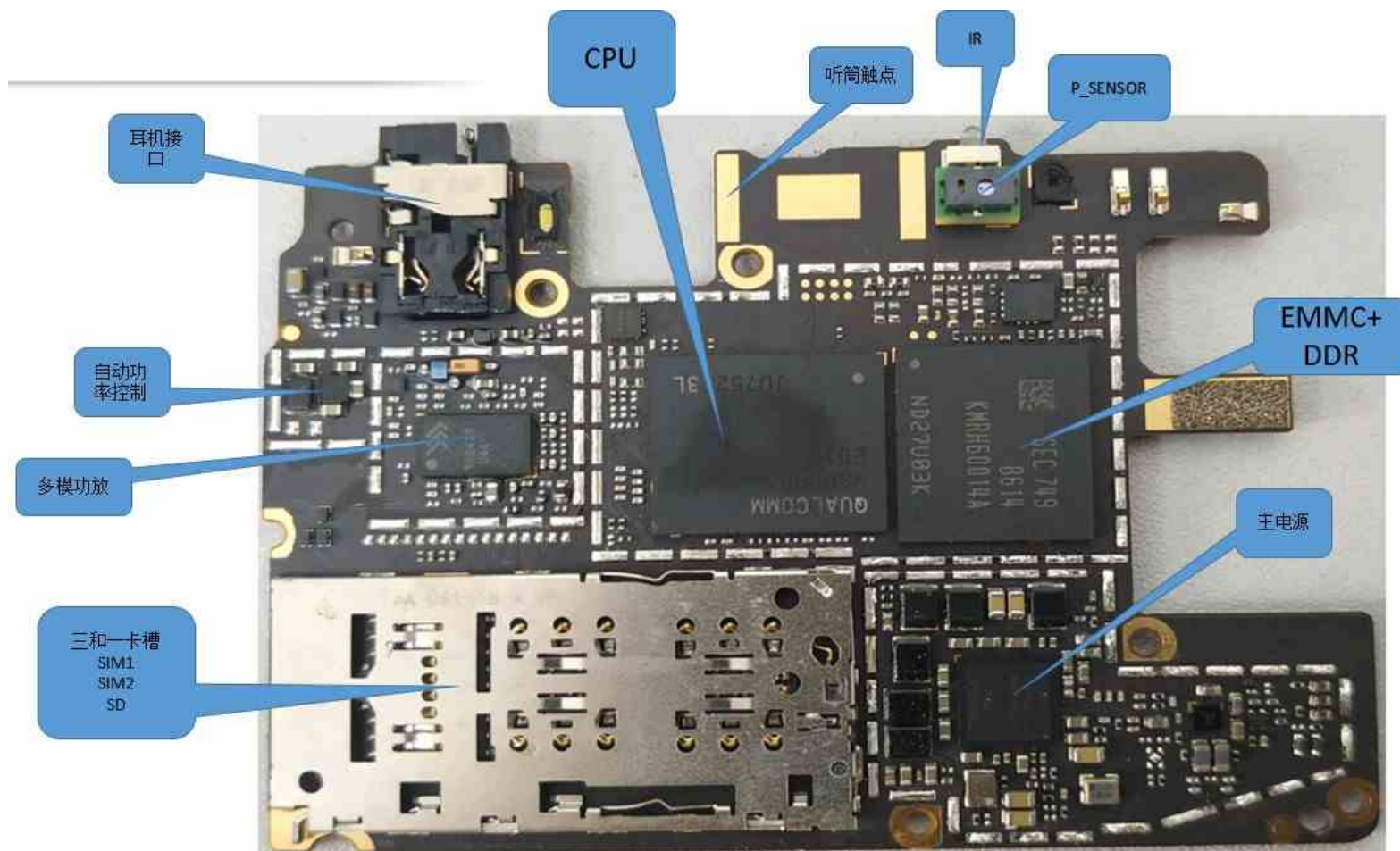
1.7 RF calibration test correlation

1. Deep brush factory software, such as brush into clear NV factory package, need to use DT tool to write FSN.
2. Open the calibration software, and connect the main receiving antenna and the auxiliary receiving two RF lines when the mobile phone is turned on. Connect the USB cable and start the RF calibration of the mobile phone.
3. After successful calibration, you need to write the flag bits with the flag bit write tool (refer to the instruction for using the flag bit tool).
When the flag bit is written successfully, it can be seen in the version information in factory mode that "PASS" is displayed in all frequency bands of RF. 4. After successful calibration, the card can be directly inserted to test the signal.

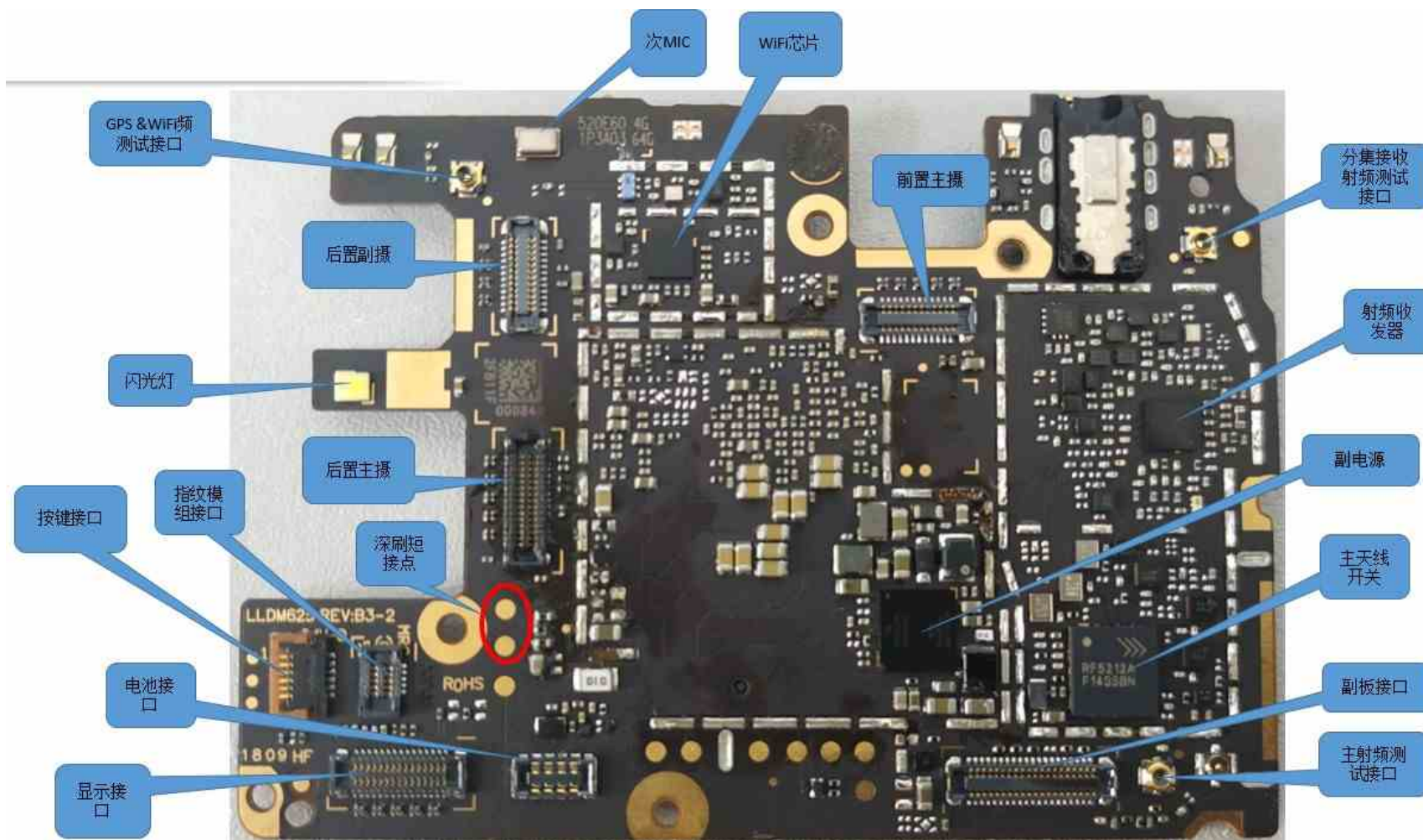
2. Brief introduction of motherboard module

2.1 Redmi S2 motherboard component distribution diagram

- TOP plane

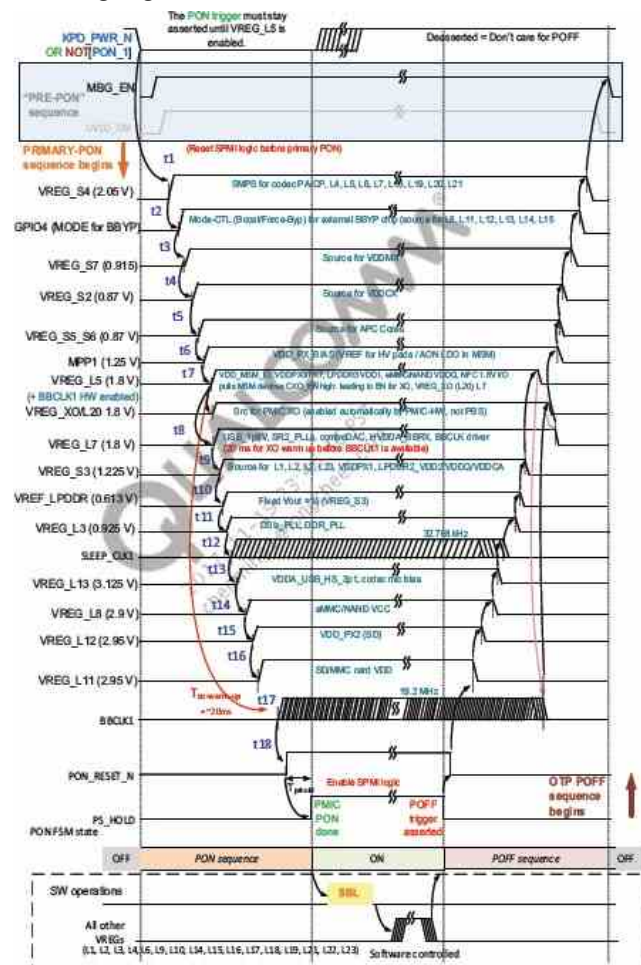


-
- BOT surface



2.2 Brief introduction of startup timing and key signal measurement table of red rice S2

Boot timing diagram:



Boot Timing:

开机时序测量表		
Symbol	测量值	测量点
KYPD_PWR_N1	1.8V	TP1502
VREG_S4_2P05	2.05V	C1610
VREG_S7_0P915	0.915V	L1607
VREG_S2_0P8625	0.8625V	L1603
VREG_S5_S6_0P8625	0.8625V	L1606
VREG_L5_1P8	1.8V	C1706
VREG_X0	1.8V	C1507
VREG_L7_1P8	1.8V	C1701
VREG_S3_1P225	1.225V	R1703
VREF_LPDDR3	0.613 V	R1905
VREG_L3_0P925	0.925 V	C1111
VREG_L13_3P075	3.125 V	C1151
VREG_L8_2P9	2.9 V	C1708
VREG_L12_VDDPX2_SDC	2.95 V	C1049
VREG_L11_2P95	2.95 V	C1711
PM_PON_RESET_N	1.8V	C501
PS_HOLD	1.8V	C1515

CPU (U0501) power supply signal and measurement points (yellow signals are also signals in startup timing):

CPU供电测量表		
Symbol	测量值	测量点
VREG_S2_0P8625	0.8625V	C1027
VREG_S3_1P225	1.225V	C1007
VREG_L12_VDDPX2_SDC	2.9V	C1049
VREG_L5_1P8	1.8V	R1003
VREG_S7_0P915	0.915V	C1002
VREG_S5_S6_0P8625	0.8625V	L1606
VREG_L3_0P925	0.925V	C1102
VREG_L7_1P8	1.8V	C1114
VREG_L13_3P075	3.075V	C1151
VREG_L14_UIM1	2.8V	C1708
VREG_L15_UIM2	3.2V	C1709
VREG_L6_1P8	1.8V	R1001
VDD_PX_BIAS_MPP_1	1.2V	C1010
VREG_L19_1P3	1.3V	R1102
VREG_S1_0P8625	0.8625V	C1132
VREG_S7_ISO	0.6V	C1140

U1901 chip bitmap:

221Ball FBGA														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	DNU	VSF	VSS_m	VDDQ_m	DAT6_m	CMD_m	RCLK_m	VSS_m	DAT0_m	DAT5_m	VDD1_m	VSS_m	VSF	DNU
B	VSF	VSS_m	VDD_m	DAT7_m	DAT3_m	VDDQ_m	VSS_m	CLK_m	VDDQ_m	DAT1_m	VSS_m	VDD_m	VDD_m	VSF
C		RSTf_m	VSS_m	VDD_m	VSS_m	DAT2_m	VDDQ_m	VSS_m	DAT4_m	VSS_m	VDDQ_m	VSS_m	VSS_m	
D		VSF	VSF	VSF	VSF	VSF	VSS_m	VDD_m						
E														
F		VSS_v	VDD1_v	VDD1_v	VDD2_v				VDDQ_v	VDD1_v	DQ29_v	DQ30_v	DQ31_v	VSSQ_v
G		ZQ0_v	ZQ1_v	VSS_v	VDD1_v				VSS_v	VDDQ_v	DQ26_v	VSSQ_v	DQ27_v	DQ28_v
H		CA9_v	VSS_v	VSSQ_v	VSS_v				VDDQ_v	DQS1_v	VSSQ_v	DQ24_v	VDDQ_v	DQ25_v
J		CA8_v	CA7_v	VSSQ_v	VDD2_v				VSSQ_v	DQS3_v	DM3_v	VDDQ_v	DQ15_v	VSSQ_v
K		VDDCA_v	CA6_v	VSSQ_v	VDD2_v				VSSQ_v	VSSQ_v	VDDQ_v	DQ13_v	VDDQ_v	DQ14_v
L		VDDQ_v	CA5_v	VSS_v	VDD2_v				VDDQ_v	VDDQ_v	VSSQ_v	DQ12_v	VSSQ_v	DQ11_v
M		VDDCA_v	VSS_v	VSS_v	VDD2_v				VSSQ_v	DQS1_v	VDDQ_v	DQ10_v	VDDQ_v	DQ9_v
N		VDDCA_v	CK_e_v	VSS_v	VDD2_v				VSS_v	DQS1_v	DM1_v	VDDQ_v	DQ8_v	VSSQ_v
P		VSSQ_v	CK1_v	VSS_v	VDD2_v				VDDQ_v	VSSQ_v	ODT_v	VDDQ_v	VSS_v	VDDQ_v
R		CKE1_v	VSS_v	VSS_v	VDD2_v				VSS_v	DQS0_v	DM0_v	VDDQ_v	DQ7_v	VSSQ_v
T		CKE0_v	CS1_e_v	VSS_v	VDD2_v				VSSQ_v	DQS0_v	VDDQ_v	DQ5_v	VDDQ_v	DQ6_v
U		VSSQ_v	CS0_e_v	VSSQ_v	VDD2_v				VDDQ_v	VDDQ_v	VSSQ_v	DQ3_v	VSSQ_v	DQ4_v
V		VDDCA_v	CA4_v	VSSQ_v	VDD2_v				VSSQ_v	VSSQ_v	VDDQ_v	DQ1_v	VDDQ_v	DQ2_v
W		CA2_v	CA3_v	VSSQ_v	VDD2_v				VSSQ_v	DQS2_v	DM2_v	VDDQ_v	DQ0_v	VSSQ_v
Y		CA0_v	CA1_v	VSS_v	VSS_v				VDDQ_v	DQS2_v	VSSQ_v	DQ23_v	VDDQ_v	DQ22_v
AA	DNU	VSS_v	VDD1_v	VSS_v	VDD1_v				VSS_v	VDDQ_v	DQ21_v	VSSQ_v	DQ20_v	DQ19_v
AB	DNU	DNU	VDD1_v	VDD1_v	VDD2_v				VDDQ_v	VDD1_v	DQ18_v	DQ17_v	DQ16_v	DNU

LPDDR3

Ground

eMMC

ODT

Power

DNU / VSF

U1301 chip bitmap:

1 VDIS_N_OUT	2 VSW_DIS_N	3 VDD_DIS_N	4 GND_DIS_P	5 VDIS_P_OUT	6 VREG_WLED	7 VDD_WLED	8 GND_WLED	9 VSW_WLED	10 MPP_3	11 HAP_PWM_IN	12 HAP_PWM_IN
13 VDIS_N_OUT	14 VSW_DIS_N	15 DIS_N_CAP_REF	16 VSW_DIS_P	17 VSW_DIS_P	18 NC	19 NC	20 WLED_SINK2	21 WLED_SINK1	22 MPP_1	23 GNDC	24 HAP_OUT_N
25 VDD_1P8_DIS_N	26 GND_DIS_N_REF	27 VDIS_N_FB	28 VDD_DIS_P	29 VDIS_P_FB	30 WLED_CABC	31 GND_WLED_I	32 WLED_SINK3	33 WLED_SINK4	34 MPP_2	35 HAP_OUT_P	36 GND_HAP
37 GNDC	38 SPMI_CLK	39 SPMI_DATA	40 NC	41 NC	42 VDD_MSM_IO	43 BIA	44 NC	45 GNDC	46 VREG_ADC_LDO	47 MPP_4	48 VDD_HAP
49 GNDC	50 CLK_IN	51 DIS_SCTRL	52 GNDC	53 DVDD_BYP	54 DVDD_BYP	55 SHDN_N	56 GNDC	57 VDD_ADC_LDO	58 REF_BYP	59 GND_REF	60 VDD_TORCH
61 CS_PLUS	62 BATT_PLUS	63 GPIO_2	64 GNDC	65 GNDC	66 GNDC	67 GNDC	68 RESIN_N	69 PS_HOLD	70 GNDC	71 FLASH_LED1	72 VDD_FLASH
73 CS_MINUS	74 BATT_MINUS	75 GPIO_1	76 GNDC	77 GNDC	78 GNDC	79 GNDC	80 USB_ID	81 WIPWR_DIV2_EN	82 GNDC	83 FLASH_LED2	84 VDD_FLASH
85 R_BIAS	86 BATT_ID	87 GNDC	88 DC_EN	89 WIPWR_RST_N	90 USB_ID_RVAL1	91 CHG_LED	92 USB_SNS	93 SYSON	94 FLASH_OUT	95 FLASH_OUT	96 FLASH_OUT
97 BATT_THERM	98 VAA_CAP	99 GNDC	100 USB_EN	101 USB_CS	102 CHG_EN	103 PGOOD_SYSOK	104 DC_SNS	105 BOOT_CAP	106 VSW_CHG	107 VSW_CHG	108 VSW_CHG
109 GNDC	110 GND_FG	111 USB_DP	112 USB_ID_RVAL2	113 CHG_OUT	114 VPH_PWR	115 GNDC	116 VSW_CHG	117 VSW_CHG	118 VSW_CHG	119 VSW_CHG	120 VSW_CHG
121 GNDC	122 GND_REF_CHG	123 USB_DM	124 CHG_VBAT_SNS	125 CHG_OUT	126 CHG_OUT	127 VPH_PWR	128 GND_CHG	129 GND_CHG	130 USB_MID	131 USB_MID	132 USB_MID
133 GNDC	134 KYPD_PWR_N	135 V_ARB	136 GNDC	137 CHG_OUT	138 VPH_PWR	139 VPH_PWR	140 WIPWR_CHG_OK	141 GND_CHG	142 USB_IN	143 USB_IN	144 USB_IN

Configurable IOs	General Housekeeping	Ground	IC-level Interfaces	Input Power Management	No Connect	Power	User Interfaces
------------------	----------------------	--------	---------------------	------------------------	------------	-------	-----------------

Figure 2-1 PMI8952 pin assignments (top view)

U1501 chip bitmap:

1 GND_XO_IS O	2 XTAL_19M_IN	3 XTAL_19M_OUT	4 VREG_XO	5 VREG_RFO LK	6 VREG_L9		7 VREG_L7	8 VREF_LPD DR	9 VREG_L1	10 GND_S4	11 VSW_S4	12 VDD_S4	13 VDD_S4
14 VCOIN	15 GND_XO_IS O	16 GND_XO	17 GND_RFCL K	18 VREG_L18	19 BB_CLK2	20 VREG_L22	21 VDD_L4_5_6_7_16_19	22 VREG_L8	23 VREG_L19	24 GND_S4	25 VSW_S4	26 VDD_S4	27 PON_RESE T_N
28 VREG_L12	29 VDD_L8_11_12_13_14_15	30 VREG_L8	31 VDD_XO_R FCLK	32 BB_CLK1	33 VDD_L9_10_17_18_22	34 VREG_L17		35 VDD_L4_5_6_7_16_19	36 VDD_L1	37 SPMI_DATA	38 VREG_S4	39 GND_S1	40 GND_S1
41 RF_CLK1	42 VREG_L11	43 RF_CLK2	44 VDD_L6_11_12_13_14_15	45 VREG_L14	46 RF_CLK3	47 VREG_L10	48 VREG_L16	49 VREG_L5	50 VREG_L4	51 SPMI_CLK	52 PS_HOLD	53 VSW_S1	54 VSW_S1
	55 BB_CLK1_E N		56 VREG_L13	57 REF_BYP	58 VREG_L15	59 VPH_PWR		60 AVDD_BYP	61 DVDD_BYP	62 VREG_S1	63 KPD_PWR_ N	64 VDD_S1	65 VDD_S1
66 LINEOUT_M	67 LINEOUT_P	68 HPH_L	69 MIC_BIAS1	70 GND_REF	71 GND	72 GND_XOADC	73 XO_THERM	74 PA_THERM	75 VREG_S2	76 RESIN_N	77 PON_1	78 GND_S2	79 GND_S2
80 VNEG_HPH	81 VDD_HPH	82 HPH_REF	83 MIC_BIAS2	84 GND_CFLT	85 GND	86 GND	87 GND	88 GND	89 SLEEP_CLK 1	90 GPIO_6		91 VSW_S2	92 VSW_S2
93 EARO_P	94 EARO_M	95 HPH_R	96 MIC1_IN_P	97 MIC1_IN_M	98 GND	99 GND	100 GND	101 CC2	102 GND	103 VREG_S7	104 GPIO_5	105 VDD_S2	106 VDD_S2
107 VDD_SPKR_ PA		108 MIC2_IN	109 HS_DET	110 MPP_3	111 OPT_1	112 MPP_1	113 VCONN	114 CC1	115 VREG_S5	116 VREF_NEG_ S5	117 GPIO_4	118 GND_S7	119 GND_S7
120 SPKR_DRV_ P	121 SPKR_DRV_ M		122 MIC3_IN	123 OPT_2	124 CSL_PWR_ N	125 MPP_2		126 VDD_L23	127 VREG_S8	128 VREF_NEG_ S6	129 GPIO_8	130 VSW_S7	131 VSW_S7
132 GND_SPKR_ PA	133 CP_VNEG	134 PDM_RX0_ DRE	135 PDM_RX0	136 PDM_SYNC	137 VREG_S3	138 VREG_L23	139 VDD_L2_3	140 VREG_L2	141 GPIO_2	142 GPIO_3	143 GPIO_7	144 VDD_S7	145 VDD_S7
146 CP_C1_M	147 GND_CP	148 PDM_RX1_ DRE	149 PDM_TX	150 PDM_CLK	151 MPP_4	152 GPIO_1	153 VREG_L3	154 VDD_S6	155 VSW_S6	156 GND_S6	157 VDD_S5	158 VSW_S5	159 GND_S5
160 CP_C1_P	161 VDD_CP	162 BOOST_SN S	163 VDD_AUDIO ID	164 PDM_RX2	165 VDD_S3	166 VSW_S3	167 GND_S3	168 VDD_S6	169 VSW_S6	170 GND_S6	171 VDD_S5	172 VSW_S5	173 GND_S5
174 GND_BOOS T	175 GND_BOOS T	176 VSW_BOOS T	177 VREG_BOO ST	178 PDM_RX1	179 VDD_S3	180 VSW_S3	181 GND_S3	182 VDD_S6	183 VSW_S6	184 GND_S6	185 VDD_S5	186 VSW_S5	187 GND_S5

Audio

Configurable I/Os

Do not connect

General housekeeping

Ground

IC-level interfaces

Input power management

Output power management

Power

Figure 2-1 PM8953 bottom pin assignments

3. Troubleshooting

3.1 Switching failure

In the process of maintenance without starting up, we should follow the principle of software before hardware, pay attention to observe whether the motherboard components are damaged, breakdown, liquid inflow, etc. In specific measurement, we should measure according to the starting time sequence.

3.1.1 Constant current without startup

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. If the brushing machine reports an error, measure whether the working conditions of U0501 and U1901 are normal (in actual maintenance, the brushing machine 11S reports an error "ACK count don't match", which generally means U1901 is damaged).
3. If the current is still constant after brushing, measure whether the startup timing and CPU working conditions are normal.

Maintenance Case 1

Failure phenomenon: Press the startup key 60mA to maintain
failure cause: U1901

Maintenance analysis: 60mA maintenance when starting up, short-circuit identification port 9008, error writing timeout when brushing, unsuccessful start of mobile phone due to poor performance of U1901, locking the problem in U1901.

Maintenance Case 2

Failure phenomenon: 70mA
maintenance failure cause:
U1901

Maintenance analysis: 70mA maintenance when starting up, short-circuit identification port 9008, successful brushing, and still failure when not starting up. Due to poor performance of U1901, the mobile phone start-up calling program is unsuccessful, so replace U1901 for fault repair.

3.1.2 No startup current is not maintained

Analysis ideas:

1. Software upgrade and troubleshooting.
2. Measure whether the startup timing signal is normal.
3. Measure whether the power supply of U0501 is normal.
4. Measure whether the USB signal circuit is normal (USB_ID; USB_DM; USB_DP, note: if these measured signals are abnormal, CPU fault can be judged from the side).

Maintenance Case 1

Failure phenomenon: 30mA
start-up is not maintained.
Failure cause: U0501

Maintenance analysis: 30mA is not maintained, the connection computer does not recognize the port, measure the short circuit of VREG_S4_2P05 to ground, remove U0501, and replace and repair it normally.

3.1.3 No startup, no current

Analysis ideas:

1. Check whether the appearance of J2702 is damaged, and if the interface is normal, measure whether the ground value of J2702 is normal (BAT_CON_ID, BAT_THERM, VBATT).
2. Power up to measure whether the VPH_PWR output is normal and whether the KYPD_PWR_N1 (TP2708) 1.8 V voltage is normal. If there is no output, replace U1301.
3. Measure whether the startup timing signal is normal.

Maintenance Case 1

Fault phenomenon: no current when starting up

Failure cause: U0501, U1501

Maintenance analysis: There is no current when starting up, measure VREG_S2_0P8625 short circuit to ground, remove CPU VREG_S2_0P8625 circuit normally, replace CPU and still have no current when starting up, measure according to starting time sequence, U1501 power supply is normal and no power supply output, replace U1501 fault repair.

3.1.4 Leakage without starting up

Maintenance analysis ideas:

1. First of all, visually check whether there are components rupture, breakdown, liquid corrosion and discoloration in the appearance of the motherboard.
2. Measure whether VBATT and VPH_PWR are short-circuited. If both VBATT and VPH_PWR are short-circuited, first find VBATT short-circuited elements, then find VPH_PWR short-circuited elements.
3. Measure whether there is short circuit in other power supply lines, and find out the faulty components according to the short circuit signal.
4. Power up to find heating elements.

Maintenance Case 1

Fault Phenomenon:
Leakage 200mA Fault
Component: U1301

Maintenance analysis: Electric leakage 200mA, measuring VPH_PWR short circuit to ground, thermal camera detecting U1301 heating, replacement and repair.

Maintenance Case 2

Fault Phenomenon:
Current Limiting
Leakage Fault
Component: C1420

Maintenance analysis: the current limiting leakage 4V voltage is not pulled down, U1301 heating is seriously removed, U1301 measurement RVPH-PWR short circuit to ground is still the same, find the fault point of this circuit component for C1420 replacement fault repair.

3.2 Restart failure

Analysis ideas:

1. Check whether the KYPD_PWR_N1 1.8 V voltage is pulled low.
2. Software upgrade, troubleshooting software.
3. Measure whether I2C to ground value and voltage are normal.
4. Measure whether the clock is normal when U0501 is powered.
5. Measure whether the motherboard has abnormal power supply caused by short circuit.
6. Replace U0501.

Maintenance Case 1

Fault phenomenon: White rice
restarts faulty component: U0501
during startup

Maintenance analysis: There is no abnormal signal in the measurement, and the mobile phone with welded CPU is turned on normally. If it is found (the green dot of CPU is marked as ball planting material), replace the CPU for repair.

Maintenance Case 2

Failure Phenomenon: Enter
Desktop Restart Failure
Mode: U3603

Maintenance analysis idea: Measure "VREG_L5_1P8" short circuit to ground, remove U3603, which is normal, replace and repair.

3.3 Crash failure

Analysis ideas:

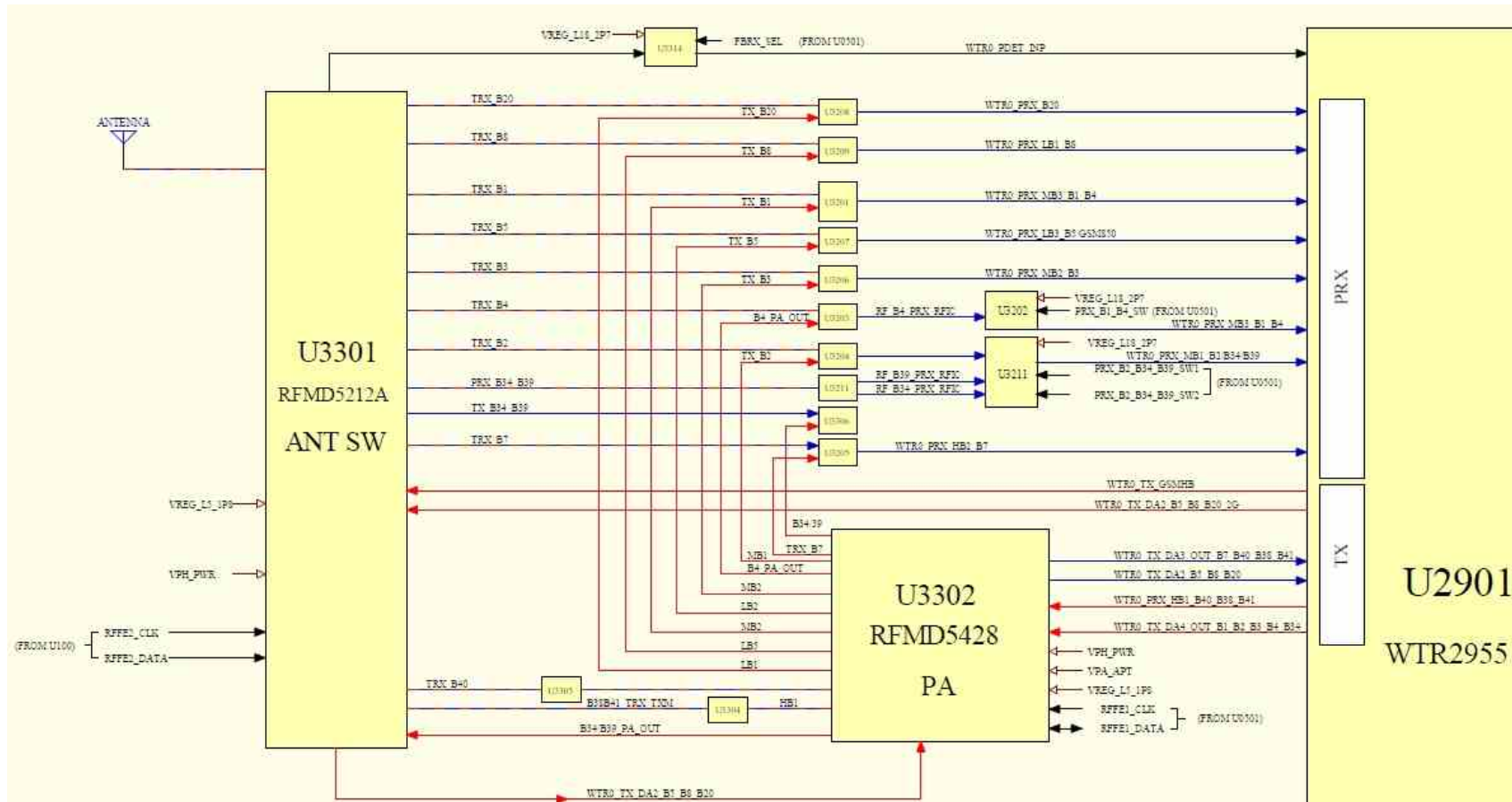
1. Software upgrade, troubleshooting software.
2. Measure whether BAT_ID and BAT_THERM are normal.
3. Measure whether the startup timing is normal.

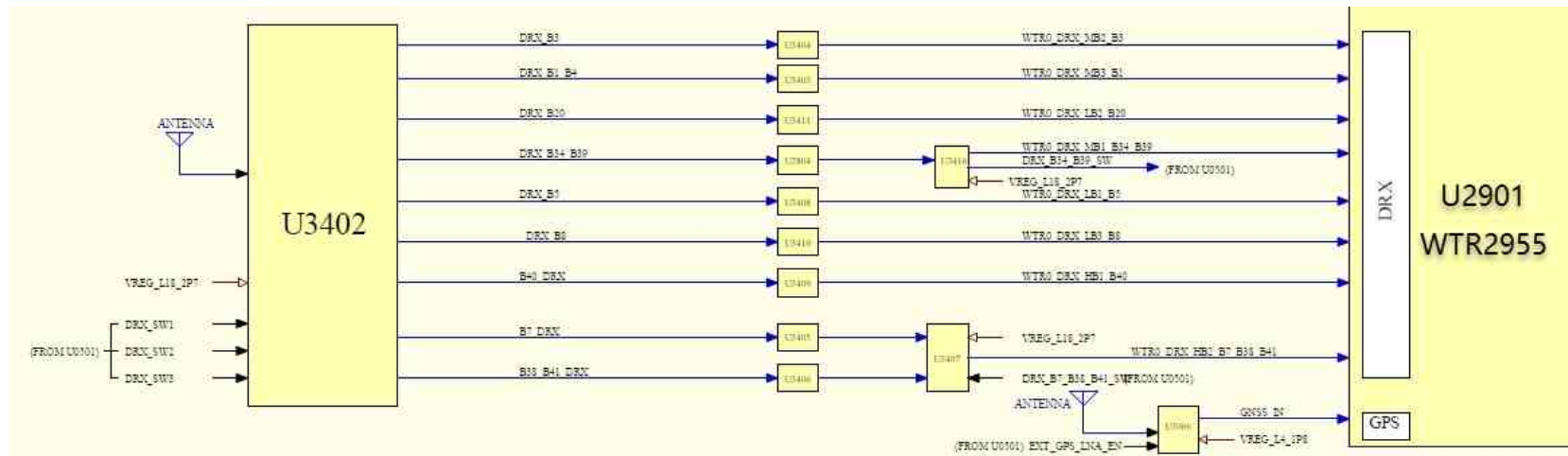
4. Measure whether the power supply of U0501 is normal.

6. Replace U0501.

3.4 Signal fault

Principle block diagram





Analysis ideas:

1. Insert SIM card to ensure normal identification, and eliminate faults that do not recognize SIM card.
2. Software upgrade, troubleshooting software.
3. RF calibration, check the specific test items through the test report. However, measure the RF path according to the corresponding system and principle block diagram to find the fault point.
4. Measure whether the power supply of RF circuit is normal and whether the RFFE CLK/DAT signal is normal.
5. If the RF calibration is normal and there is still no signal, check the SIM card circuit and whether the line from J2700 to J2701 is normal.

Maintenance Case 1

Fault Phenomenon:

No Signal Fault

Cause: U0501

Maintenance analysis: Check that the baseband information of the mobile phone is normal, SIM card can be identified, the mobile phone can't call out without signal, the welding CPU signal is normal, the RF calibration passes, and replace U0501 for fault repair.

Maintenance Case 2

Fault Phenomenon:

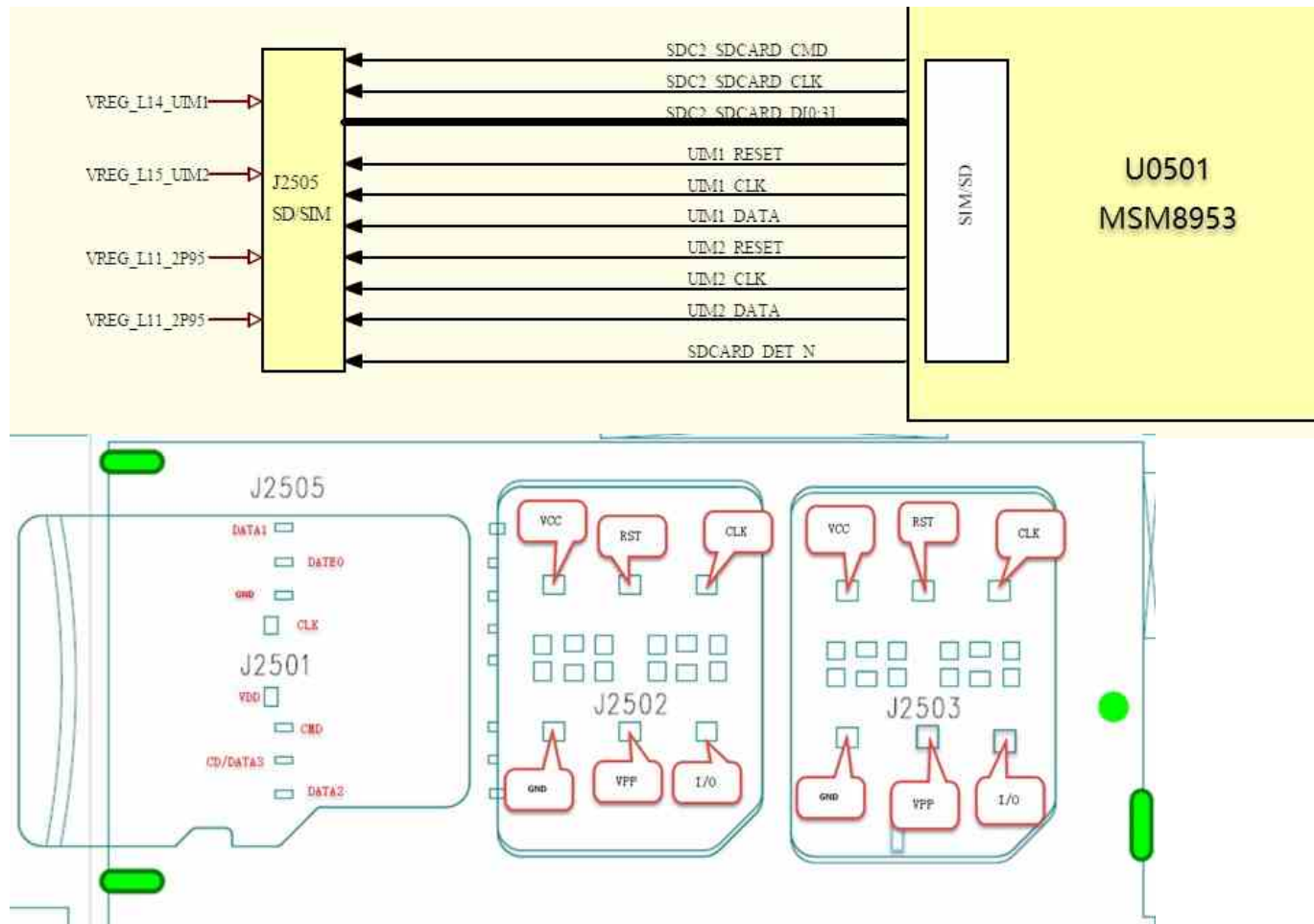
No Signal Fault

Component: U3302

Maintenance analysis: The mobile phone has no signal, the RF calibration goes offline "power supply", the short circuit of RFFE1_CLK to ground is measured, the shielding frame of SH2306 is removed, and it is found that U3501 and U3302 are burned and scrapped.

3.5 SIM card failure

Principle block diagram



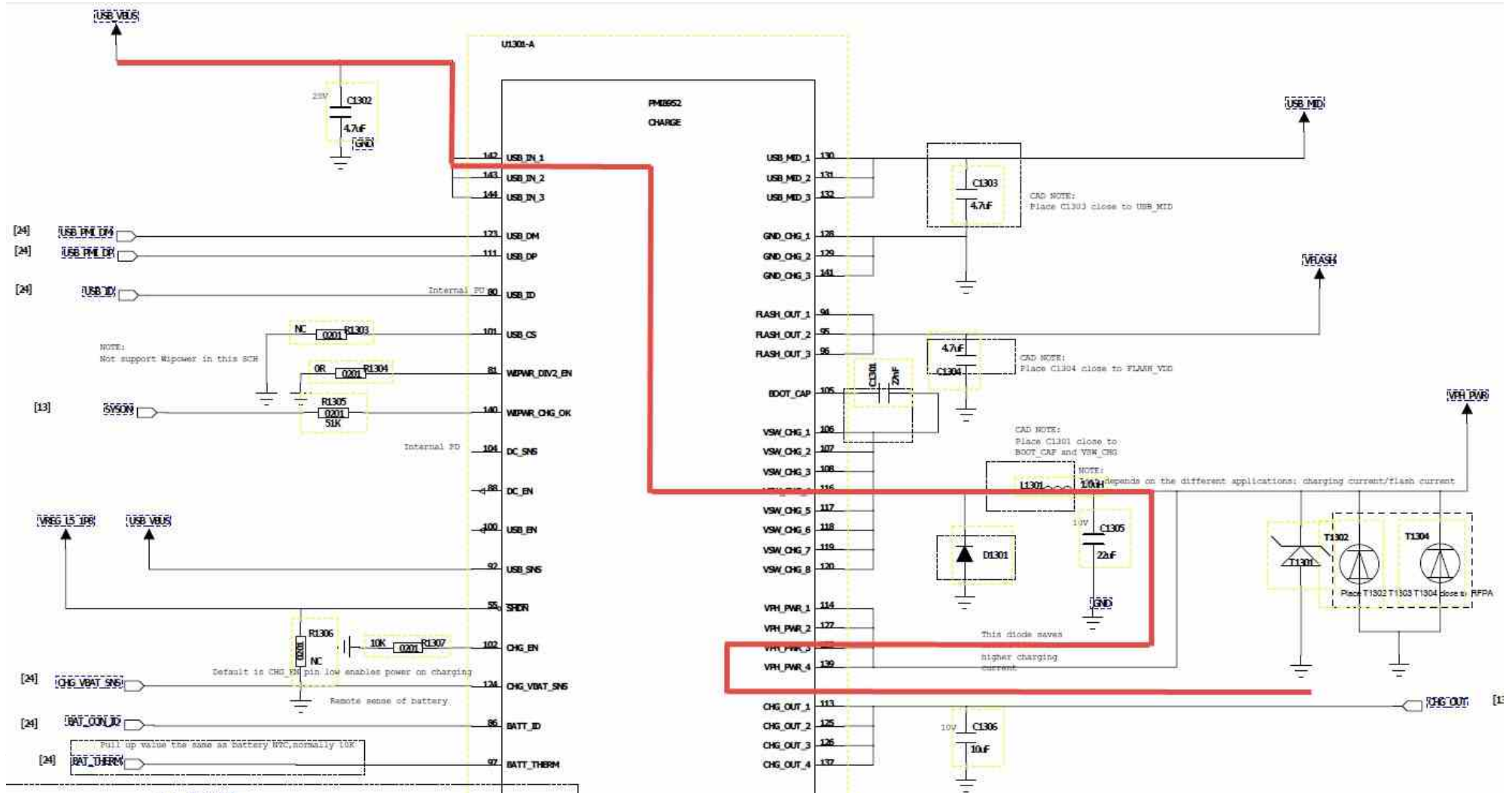
SIM/SD测量表		
Symbol	测量值	测量点
VCC	650	SIM1
RST	430	
CLK	430	
VPP	1	
I/O	430	
VCC	650	SIM2
RST	430	
CLK	430	
VPP	1	
I/O	430	
DATA1	420	SD
DATE0	420	
CLK	680	
VDD	420	
CMD	420	
CD/DATA3	420	
DATA2	420	

Maintenance analysis ideas:

- 1.First, check whether the baseband version of the mobile phone is normal. If the baseband information is normal, it is the SIM card related function failure.
- 2.Check whether the SIM card pin is deformed, oxidized or broken, and carefully observe whether there is virtual welding phenomenon in the solder joint of SIM card holder.
- 3.Measure whether the SIM card aims at the ground value normally.
- 4.Start up and test whether the power supply, clock and reset of SIM card are normal, and whether the data voltage jump is normal
- 5.If the above signal is normal, replace U0501.

3.6 Charging failure

Charging schematic diagram:



2. Measure whether the VBAT, BAT_CON_ID and BAT_THERM of J2300 are normal to ground.

3. To ensure that the above signals are normal, consider replacing U1301.

Maintenance Case 1

Fault Phenomenon:

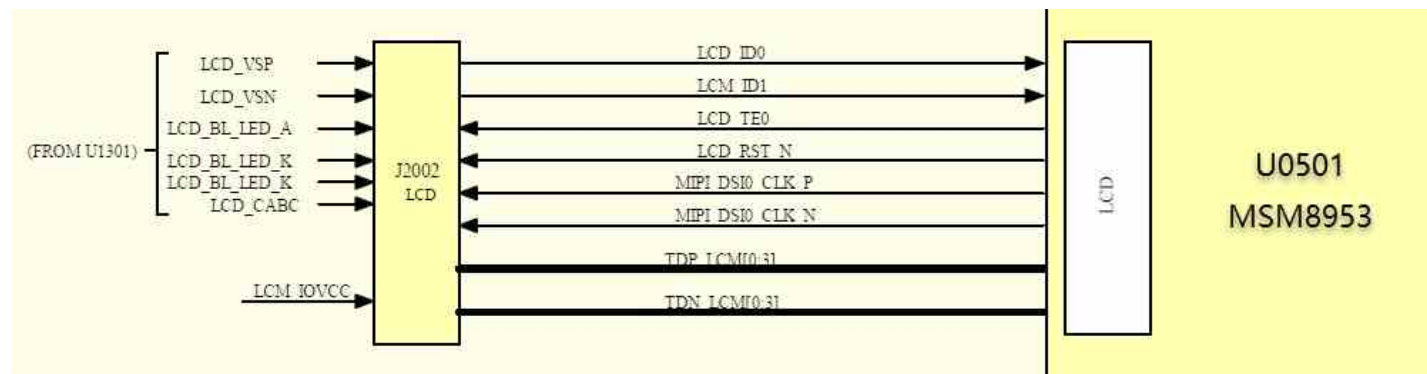
No Charge Faulty

Component: U1301

Maintenance analysis: There is no charging display on the vibration screen when the mobile phone is inserted into the charging line, and the charging current is 20-30mA. The fault point is the abnormal replacement and repair of the charging chip U1301.

3.7 Display fault

Principle block diagram



Display interface pin-to-pin measurement:

GND	580	580	390	480	1	1	738	1	444	960	960	437	490	447	GND	GND
31	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	33
J2002																
32	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	34
GND	295	295	GND	295	295	GND	295	295	GND	295	295	GND	295	295	GND	GND

Display some measurement tables:

显示电压测量表		
Symbol	测量值	测量点
LCD_VSN	-5.5V	C2018
LCD_VSP	5.5V	C2019
LCM_IOVCC	1.8V	R2005
LCD_CABC	1.8V(屏点亮)	R2022
LCD_BL_LED_A	4V-29V	C2012
LED1-	0.6V	C2013
LED2-	0.6V	C2014

Maintenance analysis ideas:

1. Visually check whether J2002 and its peripheral components are damaged or falsely welded.
2. Brush the machine to eliminate software faults.
3. Measure whether the ground value of each pin of J2002 is normal with multimeter diode gear.
4. If the ground value is normal, measure whether the power supply and control signals in the "display meter" are normal (the voltage of LCD_BL_LED_A can also be measured at 29V without adding a screen, and it is 4V in standby).
5. Replace U0501.

Maintenance Case 1

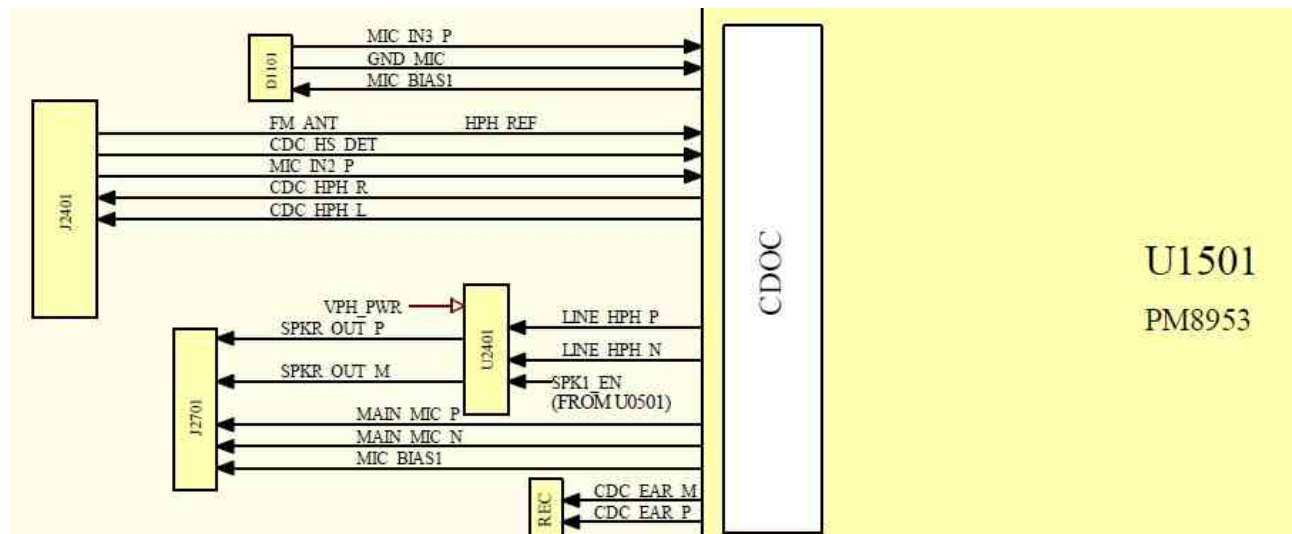
Fault phenomenon: black screen

Faulty components: U1301, C1414

Maintenance analysis: There are traces of liquid inlet in the motherboard, no display boosting in measurement, U1301 is replaced, the backlight is normal and no display, LCD_VSN is short-circuited to ground in measurement, C1414 is removed, and it is normal to replace and repair.

3.8 Audio fault

Principle block diagram



the following modules.

3.8.1 Loudspeaker failure

Speaker is connected to the motherboard through FPC. Its principle is to go to CODEC through CPU first, and then amplify and output to interface J2701 through U2401 audio power amplifier and then to speaker. SPK meter:

SPK电压测量表		
Symbol	测量值	测量点
SPKR_OUT_P	1.2V	R2405
SPKR_OUT_M	1.2V	R2406
VPH_PWR	4V	C2415

Maintenance analysis ideas:

- 1 Visually check whether J2701 looks good.
2. Software upgrade troubleshooting software.
- 3.Measure whether SPK + and SPK-to-ground values are normal with multimeter diode gear.
- 4.Measure whether U2401 voltage is normal.
- 5.If the above signals are normal, consider whether the bus from CODEC to CPU is normal.

3.8.2 MIC fault

Redmi S2 motherboard contains two MIC circuits, main/auxiliary MIC, and the main MIC is lead-type and welded on the auxiliary board. MIC meter:

MIC电压测量表		
Symbol	测量值	测量点
MAIN_MIC_P	1.2V	R2007
MAIN_MIC_N	1.2V	R2008
MIC_BIAS1	2.1V	C1816
MIC_IN3_P	1V	B2411
GND_MIC	GND	B2412
MIC_BIAS2	1.8V	C1818
MIC_IN2_P	1.2V	B2403

Maintenance analysis ideas:

1. Visually check whether J2701 looks good.
2. Measure whether MAIN_MIC_P, MAIN_MIC_N and MIC_BIAS1 are normal to ground.
3. Measure whether the MIC_BIAS1 voltage is normal.
4. If the above signals are normal, consider whether the bus between CODEC and CPU is normal.

3.8.3 Receiver failure

Maintenance analysis ideas:

1. Visually check whether B2405 and B2406 have good appearance.
2. Software upgrade and troubleshooting.
3. Measure whether REC2402 and REC2403 are normal to ground.
4. According to the audio signal trend of REC measured in the circuit diagram, reverse analysis is carried out.

3.8.4 Headphone failure

Maintenance analysis ideas:

1. Check whether the earphone interface contacts are deformed and whether there are foreign bodies in the earphone interface.
2. Software upgrade and troubleshooting.
3. Measure whether the earphone interface to the ground value is normal, do not insert the earphone when starting up, and measure whether the DET detection signal has 1.8 V voltage.
4. Repair the headphone channel according to the principle block diagram, and replace U1501 normally.
5. Replace U0501.

Fault Component:
FV2403

3.9 WIFI/BT/FM/GPS Fault

[illegible]

Bluetooth, WIFI meter:

U3603电压测量表		
Symbol	测量值	测量点
VREG_L5_1P8	1.8V	C3616
VDD_DIG_1P2	1.2V	C3609
WL_VDD_TXBB_1P3	1.3V	C3604
VDD_WL_L0_1P3	1.3V	C3622
VREG_L9_3P3	3.3V	C3613
VREG_L19_1P3	1.2V	C3620
WCN_XO_CLK	48MHz	C3611

Maintenance analysis ideas:

1. Software upgrade and troubleshooting.
3. Measure whether the power supply, clock and enable signals of U3603 are normal.
4. Take off U3603 and measure whether the bus between U0501 is normal. If it is normal, replace U3603.
5. Replace U0501.

Maintenance Case 1

Fault phenomenon: WiFi

signal weak fault cause:

U3603

Maintenance analysis: WiFi switch is normal, signal is weak, measured WiFi chip power supply voltage is normal, replace WiFi chip U3603 fault repair.

Maintenance Case 2

Failure phenomenon: WiFi

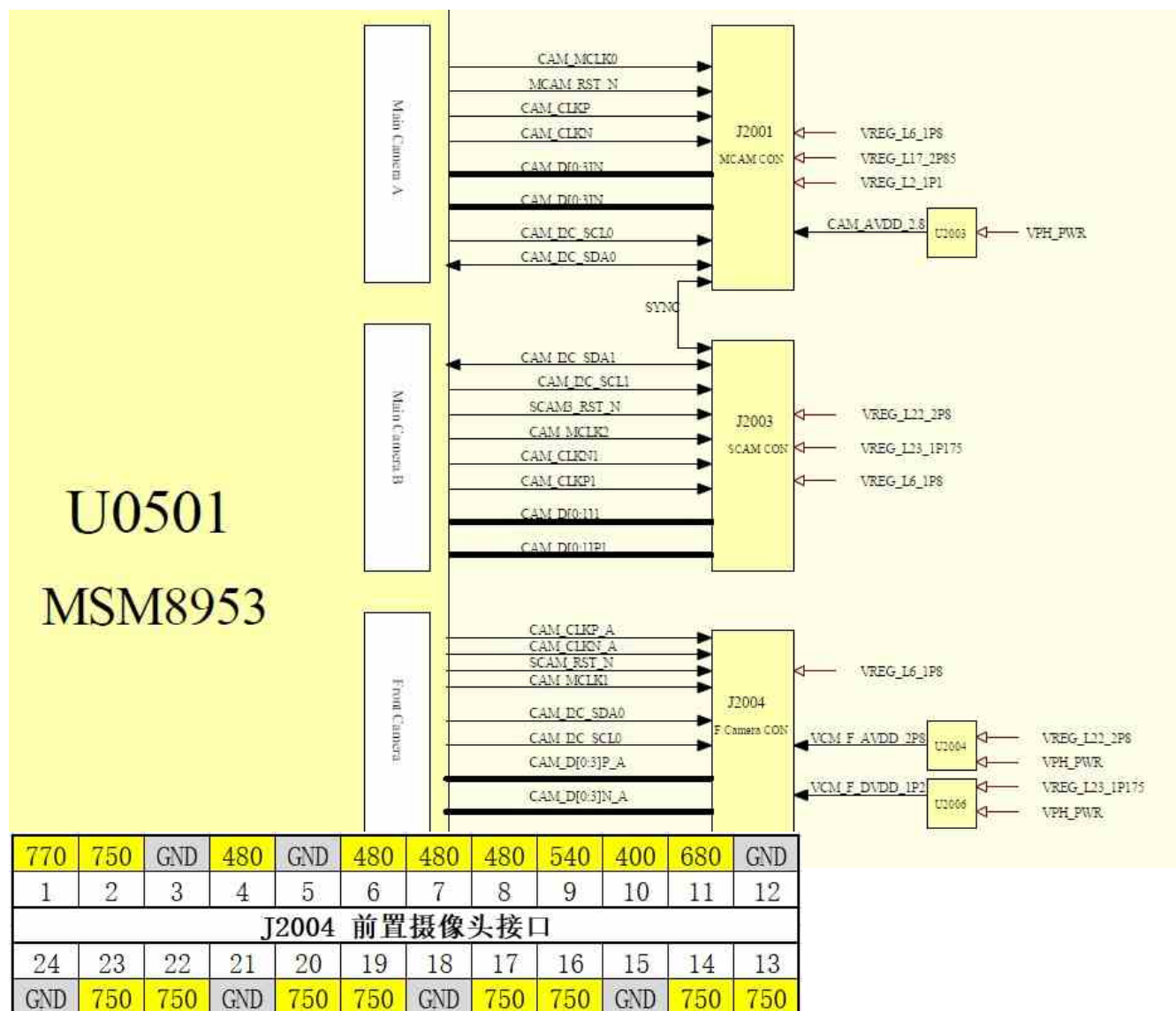
cannot be turned on Failure

cause: U0501

Maintenance analysis: WiFi cannot be turned on, Bluetooth is normal, measuring WiFi chip power supply is normal, replacing U3603 is invalid, removing U3603 measuring "WLAN_BB_Q_P" signal Open, and replacing U0501 for repair.

3.10 Camera failure

Principle block diagram



GND	480	GND	680	GND	480	1	480	480	GND	480	GND
1	2	3	4	5	6	7	8	9	10	11	12
J2003 后置副摄像头接口											
24	23	22	21	20	19	18	17	16	15	14	13
GND	750	750	GND	750	750	GND	750	750	GND	400	GND

670	GND	480	GND	550	GND	1	480	GND	480	480	GND	420	GND	400
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
J2001 后置主摄像头机接口														
30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
GND	750	750	GND	750	750	GND	750	750	GND	750	750	GND	750	750

Front camera signal measuring table:

Front Camera电压测量表		
Symbol	测量值	测量点
VCM_F_AVDD_2P8	2.8V	R2025
VCM_F_DVDD_1P2	1.2V	R2026
VREG_L6_1P8	1.8V	C2008
SCAM_RST_N	1.8V	TP2003
CAM_I2C_SCL0	1.8V	R0709
CAM_I2C_SDA0	1.8V	TP2002
CAM_MCLK1	1V	B2005
CAM_CLKP(N)_A	760	EMI2009
CAM_D(0-3)P_A	760	EMI2010/EMI2013

Rear Main Camera B signal measurement:

Main Camera B电压测量表		
Symbol	测量值	测量点
VREG_L22_2P8	2.8V	R2025
VREG_L23_1P175	1.75V	R2004
REG_L6_1P8	1.8V	C2008
CAM_I2C_SCL1	1.8V	R0710
CAM_I2C_SDA1	1.8V	R0709
SCAM3_RST_N	1.8V	TP2002
CAM_MCLK2	1V	B2004
SYNC	1.8V	R2008
CAM_CLKN(P) 1	760	EMI2006
CAM_D(0-1)N/P	760	EMI2007/EMI2008

Main Camera A电压测量表		
Symbol	测量值	测量点
VREG_L17_2P85	2.85V	C2003
CAM_AVDD_2.8	2.8V	C2025
VREG_L6_1P8	1.8V	C2036
VREG_L2_1P1	1.1V	C2001
CAM_I2C_SCL0	1.8V	R0708
CAM_I2C_SDA0	1.8V	R0707
MCAM_RST_N	1.8V	TP2001
CAM_MCLK0_0	1V	B2016
SYNC	1.8V	R2008
CAM_D(0-3)N/P	760	EMI2002/EMI2005

1. Software upgrade, troubleshooting software.
2. Detect J2001, J2003, J2004 and surrounding components for loss and damage.
3. Go to CIT to test the front camera and the rear camera to distinguish faults.
4. Measure whether the ground values of J2001, J2003 and J2004 are normal.

5. Measure whether the camera power supply, clock and reset signal output are normal.
6. Measure whether the I2C and MIPI bus output by U0501 are normal.

Maintenance Case 1

Fault Phenomenon: Front

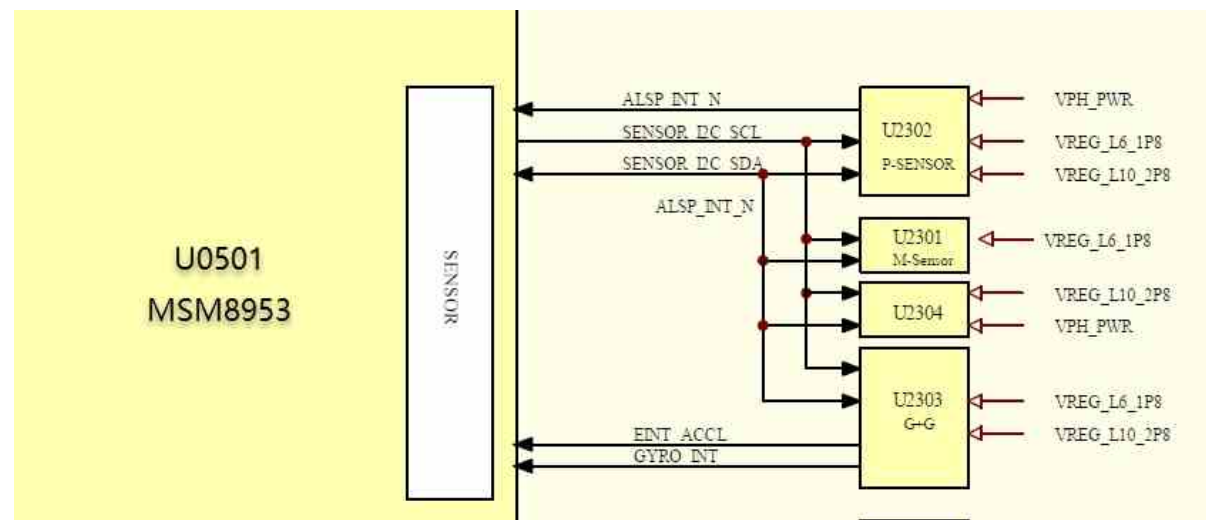
Camera Can't Turn on Faulty

Component: U0501

Maintenance analysis: The forward camera can't be opened, the motherboard has traces of liquid inlet, the SCAM_RST_N short circuit to ground is measured, the J2004 short circuit is removed, the CPU is removed, the SCAM_RST_N signal is normal, and the CPU is replaced for fault repair.

3.11 Inductor failure

Principle block diagram

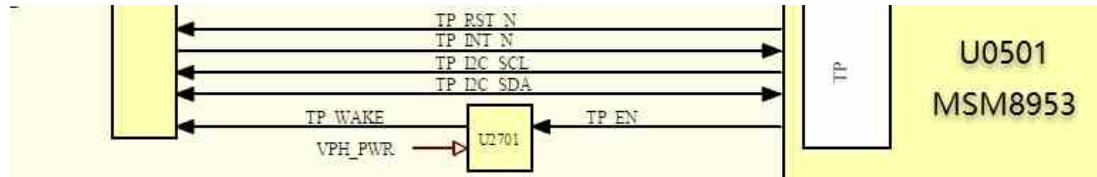


Sensor电压测量		
Symbol	测量值	测量点
SENSOR_I2C_SCL	1.8V	R0706
SENSOR_I2C_SDA	1.8V	R0705
VREG_L6_1P8	1.8V	C2305
ALSP_INT_N	1.8V	R2310
VREG_L10_2P8	2.8V	C2307

Maintenance analysis ideas:

1. Software upgrade and troubleshooting.
2. Measure whether the working conditions of the corresponding sensors are normal.
3. Replace the corresponding sensor.
4. Replace U0501.

3.12 Touch screen failure



Touch screen voltage meter

TOUCH电压测量表		
Symbol	测量值	测量点
TP_I2C_SCL	1.8V	R0712
TP_I2C_SDA	1.8V	R0711
TP_VDD	2.8V	R2702
TP_IOVDD	1.8V	R2701

Maintenance analysis ideas.

1. Check J2701 and surrounding components for damage.
2. Software upgrade, troubleshooting software.
3. Measure whether the working conditions are normal.
4. Replace U0501.

Maintenance Case 1

Fault phenomenon:

touch screen failure

Fault component:

R2702

Maintenance analysis: The software upgrade is invalid. Measure TP_VDD without voltage output. Check the fault repair of R2702 welding and welding.

Maintenance Case 2

Fault phenomenon:

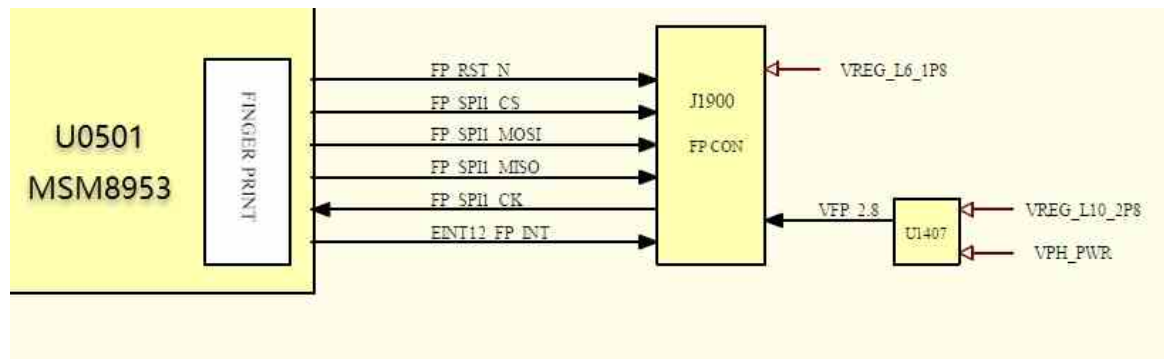
touch screen failure

Fault component:

C2709

Maintenance analysis: Measure short circuit of VREG_L10_2P8 to ground, remove C2709 for repair.

3.13 Fingerprint identification failure



Fingerprint voltage meter:

指纹电压测量表		
Symbol	测量值	测量点
VREG_L6_1P8	1.8V	T2304
VFP_2.8	2.8V	T2311
FP_RST_N	1.8V	T2305

Maintenance analysis ideas:

1. Visually inspect whether J2301 and its surrounding components are damaged. If damaged, please replace them.
2. Brush the machine to eliminate software faults.
3. Measure whether the above voltage and other signals are normal.
4. If all the above signals are normal, replace U0501.